

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

Question 1: Answer all the questions. (12 Marks)

- Observe the circuits in **Figure 1a** and **Figure 1b** and answer the questions below: [4+4+2+2]
 i) Find the Thevenin equivalent of the circuit in **Figure 1a** with respect to a-b CO3
 ii) Find the Thevenin equivalent of the circuit in **Figure 1b** with respect to c-d
 iii) If you want to run a water heater that needs **5W** power, **which circuit** should You use it, and why?
 iv) You need to **charge a battery**, but it cannot tolerate more than 12V. **Which circuit** Should you use it and why?

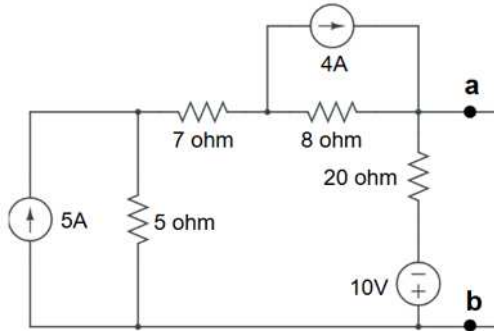


Figure 1a

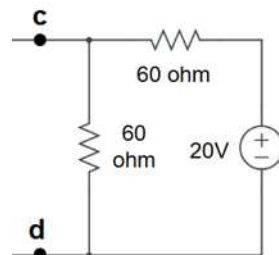


Figure 1b

Question 2: Answer all the questions. (8 Marks)

- Answer the following questions for the circuit shown in **Figure 2**: [5+3]
 i) **Determine** the branch current **i** , **I_x** , and power absorbed by the dependent voltage source using the **Superposition Theorem**. CO3
 ii) **Verify the validity** of the **Superposition Theorem** for branch current **i** by **directly solving** the circuit using **Mesh Analysis**.

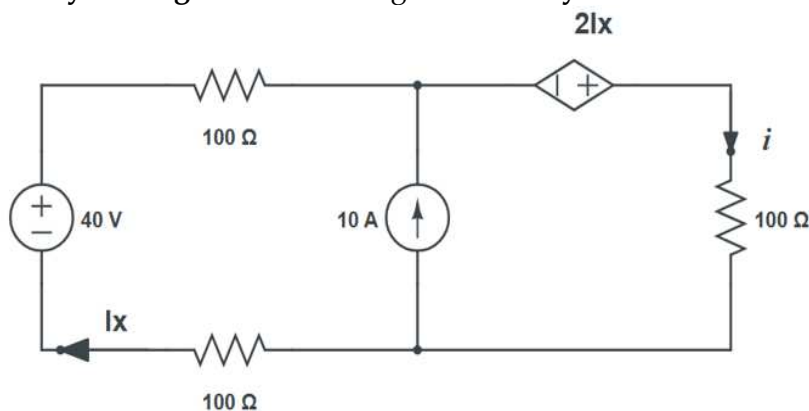


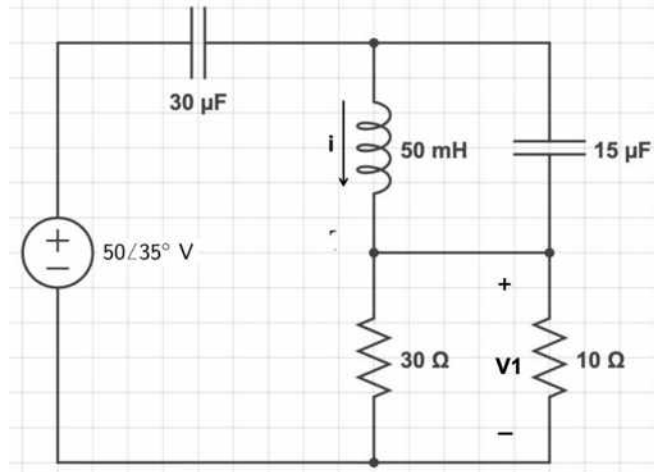
Figure 2

Question 3: Answer all the questions**(12 Marks)**

In 'Figure 3', the circuit operates at a frequency of $f = \frac{2000}{2\pi}$ Hz. Now, answer the following questions: [2+4+4+2]

CO4

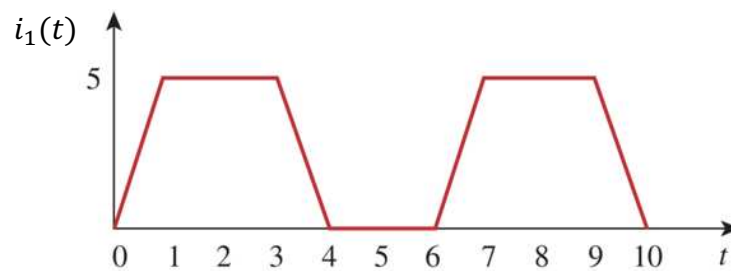
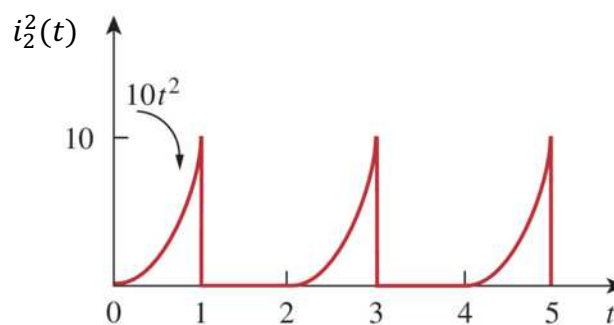
- Draw the **phasor-domain circuit**.
- Determine the **equivalent impedance** of this circuit.
- Find the **time-domain** expressions for $v_1(t)$ and $i(t)$.
- Determine the average real power absorbed by the 30 ohm resistor.

**Figure 3****Question 4: Answer all the questions.****(8 Marks)**

- Find the **rms value** of $i_1(t)$ from **Figure 4a**.
- Compare the **effective power** of the currents in **Figure 4a** and **4b** if the currents flow through a resistance of $6\ \Omega$. Which current results provide the highest power to the resistor?

[4+4]

CO4

**Figure 4a****Figure 4b**